Aditya Ingle CSE-AIDS PANEL-D Batch-D.

MPJ - Lab - 04

* Aim

Write a Java program to showcase use of file I/O Exception handling in Java

* Objective

→ To study I/O stream classes, diff types of exceptions and exception handling using various Exception classes

* Theory

→ Java I/O

It provides Input output streams to read/write data

Two types of streams

Byte stream →

FileInputStream, fileOutputStream

Character Stream →

filereader, filewriter

Used for file handling & data storage

→ Exception handling

It is used to handle runtime errors so program doesn't crash

Keywords

- try → code block
- catch → handle exception
- finally → allows execution.

- Throw → manually throw exception
- throws → declare exception

→ Types of Exceptions

- Checked Exceptions -
  Compile time (IOException)

To

- Unchecked Exceptions -
  runtime (Arithmetic exception, NullPointerException)

* FAQ.

Q5

Q1) Diff btw error and exception

→

Error	Exception
- Serious prob in system	- Prob during prog execution
- Cannot be handled easily	- easily handled (try-catch)
- System failure, memory issues	- Logical / runtime issue

Q2) Eg of checked & unchecked exceptions?

→

Checked Exception → Checked at compile time
eg. IOException, SQLException

Unchecked Exception → Checked at runtime
eg. Arithmetic Exception, NULL Point Exception

Q3) Different keywords of Exception handling in Java?

→ try - Block where risky code is written

→ catch - handle exceptions

→ finally - Execute always

→ throw - Used to manually throw exception

→ throws - Declares exception in method

Q4) Diff b/w throw & throws

→

Throw	Throws
- Inside method	- In method declaration
- To throw exception manually	- to declare exception
- Throws single exception	- Can declare multiple exceptions

Q5) Byte V/s Character Streams

→

Byte Stream	Character Stream
- Handle binary data	- handle text data
- 8-bit bytes	- 16 bit Unicode characters
- FileInputStream, FileOutputStream	- FileReader, FileWriter
- Use case :- img, audio, files	- Use Case : text files, strings